

11

Visioning a New Way to Work

Rapid CD Process	Lightning Fast	Lightning Fast +	Focused Rapid CD
Wall Walk and Visions	✓	✓	✓

After you create an affinity, consolidate the work models, and analyze them for the issues and design implications, you are ready to create the vision of how your new or redesigned system will better support your users' work or lives. Remember, tools or systems are meant to support real life activities. The elements of these activities are represented within your affinity, your sequences, and your personas. A good system fits well within, extends, and supports the activities of life. So don't start your redesign by brainstorming function—start your redesign by telling the story of what the new life will be like if you introduce new technology. This is the essence of visioning.

We recommend visioning for any Rapid CD process. Visioning can be done in one to two days and will help you conceptualize a bigger picture of what your product can do far more than simply brainstorming function or starting to sketch UI ideas. Visioning encourages you to think more systemically about your redesign—it focuses you on fitting technology into the overall intents and flow of the work process your system is situated within. This ensures that your products and systems will consider all the relevant elements of work and not break the existing work practice.

Visioning is both a “grounded brainstorm” and a storytelling session. It is grounded because the team walks the wall and collects data before they start imagining future function. It is a brainstorm because within the needs of the customer data the team members are free to present any idea without evaluation. It is storytelling because it is a process wherein the team weaves a story of how their users (which they may have

characterized by personas) address the issues uncovered in the affinity to accomplish the tasks described in the consolidated sequences.

Visioning is a method to lead groups in future scenario building, grounded in user data. Like the artist who first sketches the whole drawing structure on a canvas before filling in the detail, the team sketches the new work practice on the flipchart without getting bogged down in the details of the user interface or implementation. Storyboarding (see Chapter 12) is where you work out those details.

The philosophy behind visioning is described in Chapter 13, "Design from Data," pg. 273, in *Contextual Design: Defining Customer-Centered Systems*.

Anyone who participates in the vision must walk the affinity and consolidated sequences before they can vision. In this chapter, we outline the process for leading a team through a visioning session.

Definition

A vision is a hand-drawn, graphical representation. It is a high-level story of the customers' new practice. The vision describes what is in the new environment and how it works as told from the user's point of view. It includes possible changes in technology, user interface function, underlying business rules, role definition, and process. It can also include help, training, and overall support and services.

Key concepts

Pluses and minuses. After each vision the team evaluates both the positive and negative aspects of each of the visions generated.

Consolidated vision. The team generates multiple visions that are all evaluated. After considering what to keep and what to change the team creates one consolidated vision, which operates as the overall guide to detailed design.

The Pen. The person who draws while the rest of the visioning team tells the vision story.

The Poker. The person who reminds the team of issues they have not yet covered in the vision during the visioning process and helps ensure that the story is complete.

Process for visioning

- ☐ Preparation
 - ☐ Prepare for visioning
 - ☐ Invite people to vision
- ☐ Run the visioning session
 - ☐ Review how to vision
 - ☐ Review issues and hot ideas
 - ☐ Define parameters for the vision
 - ☐ Select a hot idea for first version

- ☐ Run the visioning session
- ☐ Evaluate the visions
- ☐ Create one consolidated vision

Prepare for visioning

Plan to do your first vision right after you walk the data. You can walk the data the afternoon before and vision the next morning or do both activities in one day. Remember that anyone who did not walk the data needs to do so before the visioning starts.

Try to hold your visioning session in the room where the affinity and sequences are displayed so that you can refer back to the data if you need to. Hang up or display the issues you generated and have the hot ideas available. The entire team should rewalk the data, rereading the design ideas if a considerable length of time has passed since the data walk.

In addition to the data you hang up on the walls you will also need to have at least one flipchart stand and pad of paper in the room. You will also need:

- Yellow 3x5 Post-it® notes, a half-pack for each person, used after visioning to put up design ideas during the evaluation
- Blue Sharpie® Fine Tip pens for everyone
- Tape to hang flipchart paper on the walls and to tape pluses and minuses to the visions
- 8½x11 or larger paper for capturing pluses and minuses

Note: Good visioning sessions need energy, especially mental energy, from all the participants. Try to avoid scheduling the session at the end of the day or other times when people aren't fresh.

Invite people to vision

The people at the visioning sessions will be the ones who generate the high-level design for your products. As such they should represent the core team, any stakeholders who wish to contribute, and anyone else who has knowledge that you need. Remember, we do not recommend that anyone participate in the visioning session without walking the data.

Visioning sessions work best when you have enough people to stimulate ideas and get multiple perspectives. Your goal is to get the people in the room who have the design skill and the technical, market, process, and business knowledge that taken together will inform the new design (see the box, **A vision is only as good as the team's combined skill**).

But, too many people in the session can make it difficult to manage: everyone will be talking at once, quiet people will have a hard time contributing, and the story may become less coherent. Plan to have no more than 10 people at the visioning session; an optimal size is between four and eight people. If you have more than eight people, run parallel sessions with each group visioning a different hot idea.



A vision is only as good as the team's combined skill

Customer data is the context that stimulates the direction of invention. But there is no invention without understanding the materials of invention: technology, design, and work practice. The visioning team needs to include people who understand the possibilities and constraints of the technology. If the team is supposed to design web pages and none of them has ever designed a web page, they will not be able to use web technology to design. When the people who always designed mainframes were told to design WYSIWYG interfaces, they replicated the mainframe interface in the WYSIWYG interface. Similarly, if the team has no interaction designers or work practice designers on it, the resulting vision will not be as powerful as if they did. This is why we recommend that design teams include people with diverse backgrounds representing all the materials of design and the different functions of the organization. Only then will an innovative design, right for the business to ship, emerge.

Review how to vision

The visioning process has three major steps. First, create three to four individual visions on flipchart pages. Each will take about 30 minutes. Second, evaluate each vision and identify what works and doesn't work about each one. These steps are continued until the team feels they have covered the key issues identified. Finally, take the good parts of each vision, overcome the bad parts, and create the final consolidated vision that will guide the team's work. Depending on the scope of the work visioning can be completed in one to two days.

Like all Contextual Design meetings, the visioning session has a process, roles, and ground rules that you'll need to review with the team before starting.

Introduce the visioning process

Explain how to vision. Tell the team that a vision is a group storytelling process in which every one participates, much like the chain ghost stories they told as kids. One person starts the story and others chime in adding to the story as it unfolds. This is the story of the users we interviewed performing the tasks we saw with the new technology and processes we are going to invent.

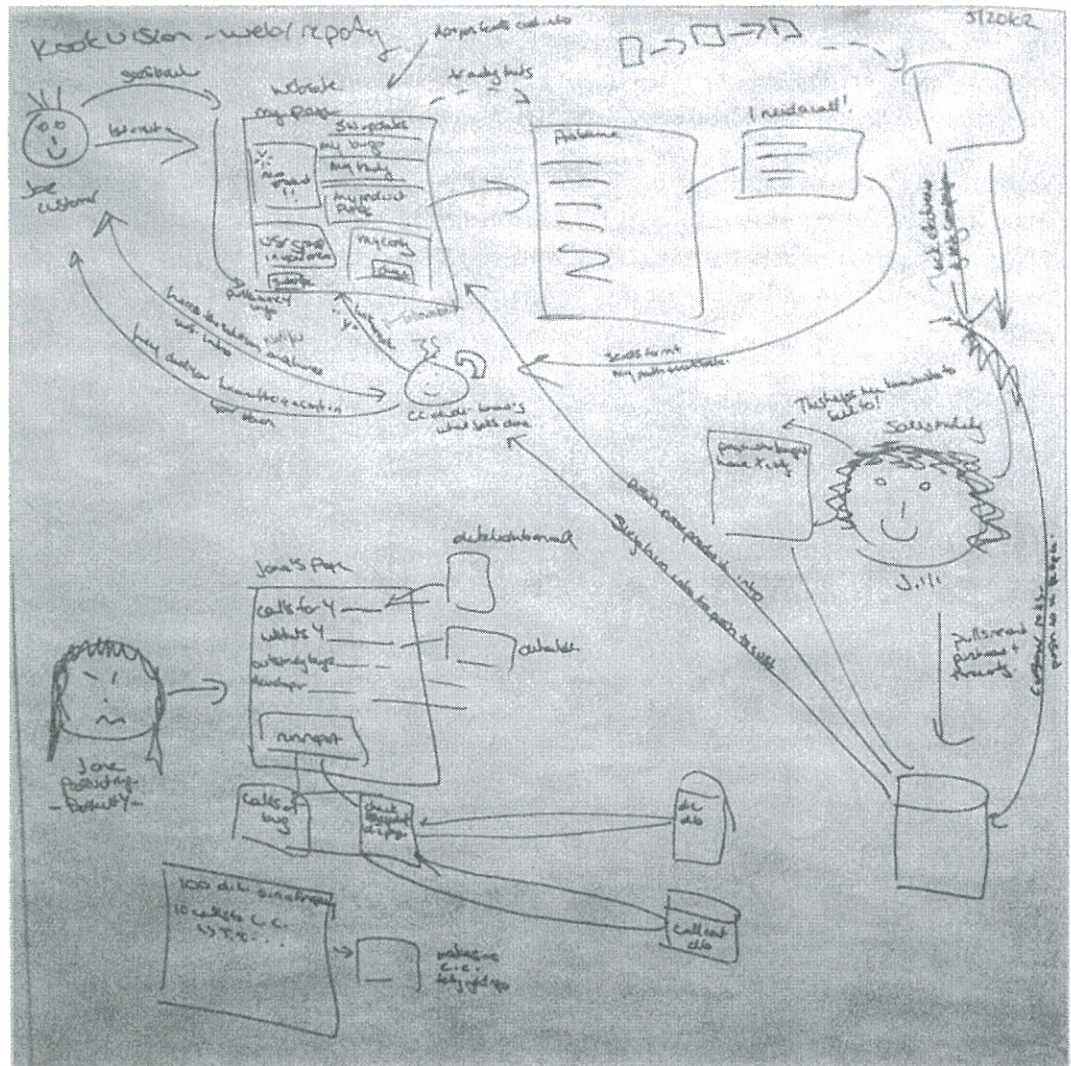
Everyone's ideas are welcome as long as they are developing the story from the point of the hot idea being developed or the task being explored. A vision includes manual processes and technical aspects. It has UI concepts and human interactions. It has automatic function and simple walking, reading, and talking. It is the story of the entire work practice or life activity that will be affected by the introduction of the technology.

Explain the perspective of the story. The vision is told as a story from the perspective of the user, not from the system or the product. Encourage this by starting the vision by asking "Who am I?" and "What am I doing?"

Describe the level of detail. The vision is high-level in that it doesn't work out the details. However, high-level does not mean "vague." The vision works out a specific set of user stories based on the tasks in the sequence models but it does not work out detailed UIs or underlying technology implementation. The vision focuses on what the user sees and does and what the system monitors and automates at the level of function, not at the level of detailed design.

Figure 11-2 illustrates that there are user-interface-like areas but they do not imply layout—only coherent collection of function. Also that they show the process of use within the workgroup and what happens under happy and upset conditions.

Figure 11-2: Apropos' web vision is an example of how to represent functions needed without worrying about the UI.



Introduce the roles

The Pen. The Pen is the person drawing the vision as the rest of the team tells the story (see Figure 11-4 for examples of vision drawings). The Pen is in service of the team, which means that they cannot contribute to the story. The Pen can ask questions for clarification, and may throw in a design concept or two. But the Pen cannot start telling the story as a major contributor. If someone feels strongly about a hot idea and wants to participate in the telling, they should not be the Pen. If they know the visioning process, an outside facilitator can act as the Pen.

The Pen has two roles: encourage people to talk, and fit their ideas into the vision as it is developing. The Pen listens for ideas that are on the story thread, postponing ideas that are too far off the main line for another vision. When an idea conflicts with the thread the team is working on, the Pen adds it to the list of hot

ideas—this keeps the thread coherent, while assuring the team member that his idea has been heard and will be dealt with later.

The Pen also moderates the rate of talk, asking people to wait if he or she needs to catch up, calling upon someone who had been trying to contribute, and generally making sure everyone's ideas are heard. The Pen is not a gatekeeper for ideas and will frustrate the team and the flow of the story if they do not draw quickly enough and include the ideas of all team members.

Figure 11-3 shows a complete workflow. To reveal what the system needs to do to support the entire process, process steps like the telephone conversation are included, communication across multiple people playing multiple roles are explored, and multiple devices (PCs and handhelds) are included.



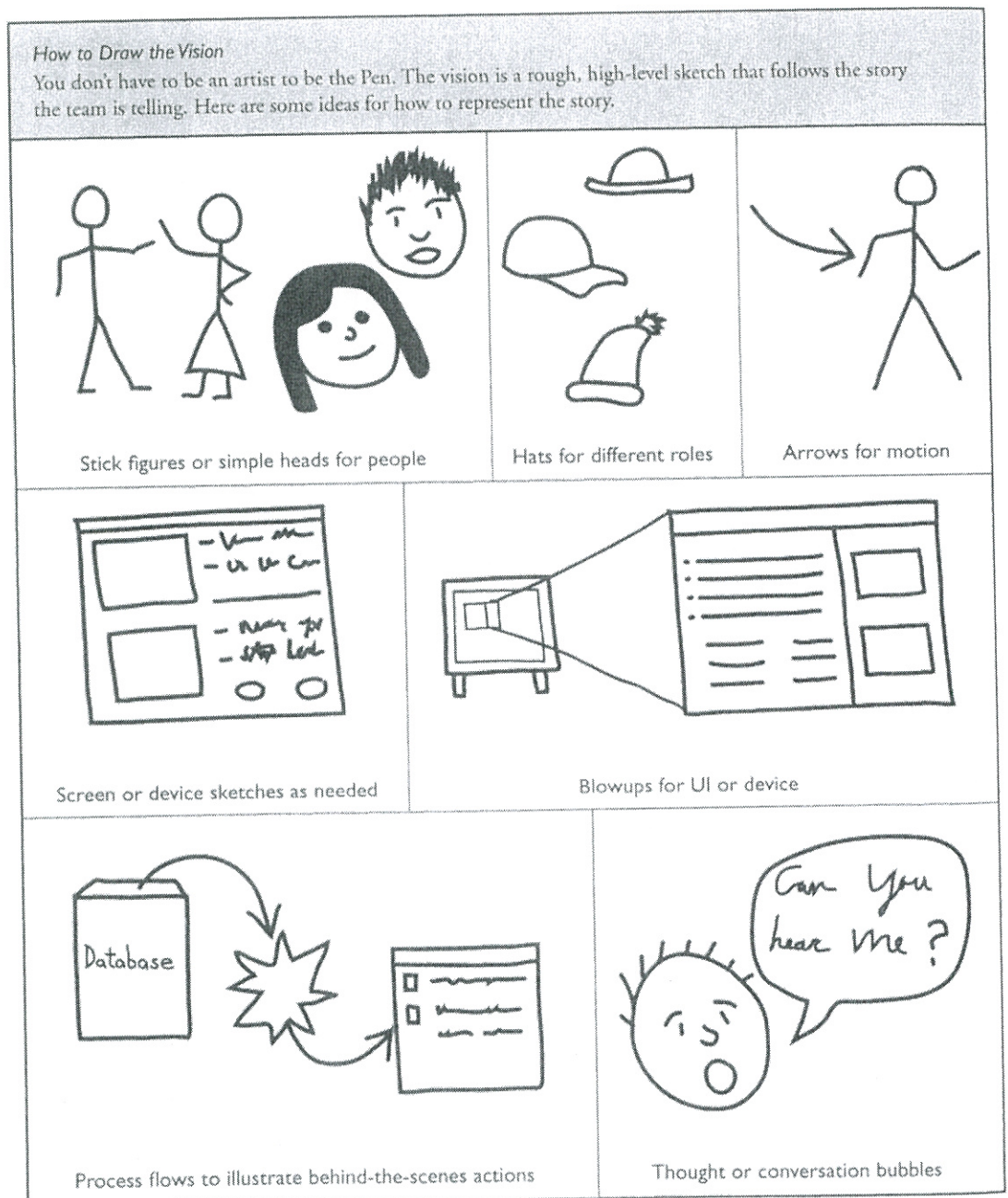
Figure 11-3: Vision of B2B purchasing across the web. The UI is envisioned in the context of the users' total workflow.

The Poker. The Poker can be anyone on the team. The Poker participates in the vision but also has an eye on the issues the team said they wanted to address. When the team reaches a lull in the storytelling the Poker will remind the team about an issue that has not yet been addressed. This will stimulate additional storytelling and lead to a more complete solution.

When moderating, this person's job is to be sure the rules are being followed, that everyone has a chance to be heard, and that no one person is dominating the story.

Participants. Except for the Pen, the entire team participates in the telling of the vision.

Figure 11-4:
Examples of how
to draw compo-
nents of the vision.



Explain the rules

- There is no evaluation of ideas as good or bad during the initial visioning. There is a separate evaluation period later.
- No idea belongs to a person; everyone participates in elaborating a group vision.
- Talk to the Pen (drawing the vision), not to each other. You are creating what is on the flipchart—if it isn't on the flipchart it won't be in the design.
- Offer design solutions and user actions—don't ask the team questions about what we want to do about this or that issue. Visioning is not a discussion, it is a time for all to cocreate the story of how things will work in the future.

Review issues and hot ideas

The intent of reviewing issues and hot ideas is to prime your brain in preparation of the vision. Prior to starting make sure you:

- Make or review the list of issues generated by walking the affinity and consolidated sequence models. You want to do this so you know where you need to take the design. You also need to know what the problem areas are so that you can correct them.
- Make or review the list of hot ideas. This list comes from the hot ideas the team generated while walking the data. It is the starting point for the vision.
- Optionally, make a list of technologies available to you. This step can be very useful because it reminds the team of the technologies available to them for invention. If anyone does not understand one of the technologies it can be explained or even demonstrated. People can design only from what they know, so be sure everyone on the team understands the technological assumptions for your project.

Define parameters for the vision

Prior to visioning the team needs to understand the parameters within which to vision. The most important of these are technological, business, and time constraints.

If *technology platforms* have been picked the team needs to know what it can and can't do. Manage this by asking a technologist to present the technology platforms to the team before visioning. Let the team ask questions so that they can understand what they can and cannot envision.

Business constraints can take many forms. The team needs to understand the business goals whether they are designing products or internal systems. They need to understand the value that the business is expecting from the system. If it is an ERP (Enterprise Resource Planning) system or an internal business system the team needs to understand any process reengineering expectation or upcoming business changes that they must consider. Ask someone from the business or from marketing to present the business goals and expectations to the team before they start visioning. Clarify how "set in stone" these decisions are to understand how widely you can vision.

Time, meaning expected timeframe until shipment, is the aspect the team most often worries about. The danger of over-focusing on time is that the team will vision so narrowly that they will not see how different parts fit into a whole solution. Remember that a broad vision can be broken into parts and developed individually. But if it was all visioned together each release is more likely to hang together for the user, particularly if you ship within a reasonable timeframe.

Our general recommendations are as follows:

- For a standard development team, use only current shipping technology and invent a vision that can be shipped in three to five years.
- For an advanced research team, use technology that will mature within five years and vision in a five- to 10-year timeframe.
- For a team that is trying to ship a release within one year but wants to develop within a bigger picture, use only current shipping technology and invent a vision that can be shipped in three to five years. But then back off the vision and do a one-year vision based on the longer term vision. Don't start with the short vision.
- For a team that only wants to identify the top things to fix in the next release, don't vision at all. List the top key fixes in order of value to the user; then work with development to determine which of these are easy and hard to implement. Create a final list balancing user need with engineering effort.
- For a team using XP, use only currently shipping technology and invent a vision that can be shipped in two years. Then break up the vision into chunks of related function and start with the chunk most central to the work. Use the vision to direct work after that, changing priorities based on the continuous user feedback you are getting.

In general, remember, a team will not feel empowered to vision more widely if they do not feel that there is a reasonable plan to identify the first release. Explain how the vision and the visioning process can be used to direct both short and longer term decisions. Discourage a tightly constrained vision; it usually doesn't work for the user because their work is not being imagined as a whole. Rather let the team vision more broadly, and then create a last vision of the next release. In that way you can get the best of both worlds.

Since visioning itself is quick, it pushes a Rapid CD process forward while it creating a shared understanding across functional groups and other people that have a stake in the ultimate solution.

Select a hot idea for the first vision

If you built a hot ideas list after walking your affinity and the consolidated models, you already have a list. If you haven't, your first step is to get your list (see Chapter 10).

Because hot ideas are listed in a brainstorming-like fashion, you may have overlapping hot ideas, or several smaller ideas that could be combined into one big idea. Take a few moments to consolidate the list and identify the top groupings of hot ideas. This also helps the team see that smaller ideas can combine to create more systemic ideas.

Example: eChalk

eChalk's hot ideas list started off as having over 30 hot ideas—big and little. This is very exciting, but also a little overwhelming. In several cases they had related ideas that were combined into one big idea.



For example, in this list the first idea—Virtual Classroom—is the hot idea. The other ideas are extensions of that one big idea:

Idea: Virtual classroom—everything to do with a class is online

Idea: Bug tracker for the students—keep track a student's progress throughout the year and use this data to generate a progress report

Idea: Student files—an area where students can store their information as they move through a school district

Idea: Centrally store all communication that relates to a student (e.g., field trip forms, student work, etc.)

Idea: eChalk pen pals

After about 10 minutes eChalk consolidated their hot idea list down to five bigger ideas that represented all the smaller ideas.

Idea: Virtual support person

Idea: My eChalk

Idea: School showcase

Idea: Online classroom

Idea: Forms life cycle

Whether you have five, 10, 20, or more hot ideas, the team needs a fast way to decide where to start. Rather than taking time to debate this, we suggest that you simply vote.

Where you start actually doesn't matter because you will do more than one vision anyway. More importantly, any vision naturally pulls in aspects of multiple hot ideas as the story unfolds. So try some fun ways to identify the first idea.

Here's the one we tend to use most often. It takes only about five minutes, and gets the team energized and excited.

1. Explain to the visioning team that they get to vote on which hot ideas they are going to vision.
2. Explain that a Post-it® flag is equivalent to one vote.
3. Pass out an equal number of Post-it® flags to each team member; six is a good number.
4. Let each person cast their votes anyway they want. They can vote for six ideas, or put all six votes on one idea, or spread them out in any other way.
5. Have everyone stand in front of the hot ideas flipchart, and remind them that they can't stand back and wait to see how the team is going to cast their votes.
6. Give the team enough time to cast all their votes. But don't let this go on too long; after three or four minutes, give them only two more minutes to cast all of their votes.

7. The hot idea with the most votes is the one you will start with.
8. Also, you can vote simply by a general applause or “noise” metric; the hot idea with the most noise wins. The goal is to eliminate debate and just get started!

Run the visioning session

The Pen steps to the flipchart and labels it with the title of the hot idea. Then the Pen asks “Who am I and what am I doing?” Once a team member starts providing ideas you are into the first vision. Watch that everyone is participating, make sure that the ideas get onto the paper, and listen to the Poker, who will remind you of areas to cover (see the box, *Managing the team*).

Listen for when someone suggests an option that would be better addressed by treating it as a separate idea. Each idea has a core theme, which you should state explicitly—if the option just extends the same theme, incorporate it. If it suggests a new theme or significantly modifies the current one, treat it as a new idea. Tell them you will elaborate the idea later. Don’t try to stretch an idea too far.

When the first vision is finished go on to a second vision. Listen for when the team stops elaborating on an idea. Move on to the next idea quickly. Spend no more than 20 to 30 minutes on each vision.

Choose the next idea to cover a new area of customer work, draw on a very different type of technology, or allow for a different level of risk. Draw each idea on a separate sheet of paper.



Managing the team

Here are some problems with participants in the visioning process you may encounter.

Worrying about the vision. Seeing an idea drawn out will disturb some people. They will see all the flaws or get stuck on how to implement it. Remind them that you have committed to nothing—it is a flipchart page. Later we will evaluate for what works and what doesn’t and change things.

Not speaking up. Make sure everyone is participating and feeling heard. Part of people not speaking up may be the number of people involved; visioning is difficult to do with more than 10 people. If necessary, break into smaller groups.

Too tight to vision. Some people or even entire teams will have a difficult time letting go and thinking creatively. To help them loosen up, have a silly break beforehand, vision while standing up, and generally help them let go physically. Remind them that thinking widely will lead to a more innovative product and that you will evaluate later. Alternatively, suggest some wild design idea to start the visioning session and push them wide. When they pull back, they’ll most likely settle in a broader place than they were in before.

Meticulous group. Some people are very methodical and need to put every piece in place before they can go to the next. When the whole group is like that, don’t try to push them faster than they want to go for the sake of time. Teams like this don’t mind putting in the time if it means they can work their ideas out.

Mixed style group. When only certain team members are detail oriented and others want to be very open and visionary without detail, talk to them about appreciating what each brings to the vision. Let the visionary set the direction and let the detail person fill in the first layer of detail, without going too deep.

If your group is big, you can run parallel visioning sessions or start together and then split up.

Do at least two visions before evaluating—three to four is better. But this also depends on the scope of the project. The goal is to cover the work practice you are designing for and to touch upon your most interesting large hot ideas. See Table 11-1 for tips on running the vision.

Note: It is critical to a successful visioning session that you have a good Pen. This person needs to be able to listen and synthesize ideas quickly, without a lot of filtering or explanation. If the Pen is slow, the meeting gets boring. If people don't feel that their ideas are being written down, they will get frustrated. Change Pens and speed up the session.

Table 11-1:
Tips for running a
productive vision-
ing session.

Dos and don'ts for the visioning process	
Don't	Do
Use multiple pages for one vision.	Keep the vision to a single flipchart page. Running over multiple pages usually indicates you are trying to address more than one coherent task or perspective in a vision. Start fresh on a new page with a focused topic.
Talk in generalities.	Say specifically who the users are and what they are doing, saying, thinking, and seeing.
Over focus on the system or the UI.	Acknowledge databases, processors, web servers, and so on that are working in the background, but don't work out the implementation. Roughly sketch out UI screens or devices to represent what the user might do there. Don't worry about buttons and navigation.
Feel constrained that this will be your final design.	Create multiple visions and evaluate them. Imagine possibilities that may be too difficult to implement. Remember that the evaluation step comes next and the vision can be scaled back once you have it.
Be prematurely constrained or narrow in visioning.	Vision as broadly as you can within the constraints of your project focus. You can then pull back once you see the possibilities and evaluate them against resources, schedules, and technical constraints.
Try to work out all the details.	Wait for storyboarding to work out the details.
Talk to each other and not to the Pen or have side conversations.	Share your ideas with everyone. Side conversations are disruptive to concentration and then your ideas aren't captured. You aren't discussing the vision with each other and asking questions about how it works, you are telling the story to the Pen.
Worry about being neat or perfect when drawing.	Don't worry too much about neatness. Visions are rough brainstorming that are done quickly because that tells a more integrated story. Trying to make it neat will slow down the storytelling and destructure the story.

**Example: eChalk**

eChalk's Online Classroom vision (see Figure 11-5) is based on the idea that core classroom tasks like scheduling class activities, making assignments, getting permission slips from parents, communicating with parents about student behavior, and taking attendance all need to be available in one web application. These activities are from the sequences that eChalk collected and consolidated.

At first glance the vision might look linear, but in reality it reflects activities that are happening throughout the year, and can happen in any order. The story of the work involves multiple people, so each individual playing a role is shown with a little head, labeled with his or her identity.

Process steps like thoughts or conversations are represented in bubbles next to the person's head. Screens or web pages are shown as nothing more than rectangles, populated with content. At this point there is no attention to what the actual UI might be—checkboxes versus list boxes, shape of buttons, and so on are not relevant to expressing what functionality is needed to support the work in the story. When the user takes an action like checking something off or clicking a button, that shows on the screen drawing.

The underlying system that is storing information and populating screens is represented by databases or cans. What the system is doing is represented at a very high-level, sometimes with just a name for the database and other times a few words or bullet points. The vision isn't implying that multiple systems or databases will be needed; this is just a way to indicate what has to happen under the hood to respond to the users' actions on the various screens.

Evaluate the vision

The purpose of evaluation is to identify what works and overcome what does not work in the visions. Evaluation is not conducted to pick among competing visions. Each vision page has processes, function, and design elements that the team will want to keep. Treating the picture as a monolithic whole makes people think they have to implement the whole thing as is.

But good design is the invention, recombination, and iteration of parts within a socio-technical system until it meets the needs of the user, the technology, and the business. Evaluating a vision by looking at its parts encourages creative redesign and keeps the team from rejecting the visions because of aspects that don't work.

Pluses and minuses, as we call it, is an informal way to help the team step back and review their ideas critically. It is through the critique that they will see what to keep, what to let go of, and how to change things that don't work so they do.

Pluses and minuses is a fast group process to raise issues and come to consensus—a process that is much better than long prioritization meetings. Do pluses and minuses with the same team that produced the visions in the first place, right after the visioning session.

An evaluation session covers about two to four visions. Do not evaluate each vision after you create it—evaluate a group of visions together.

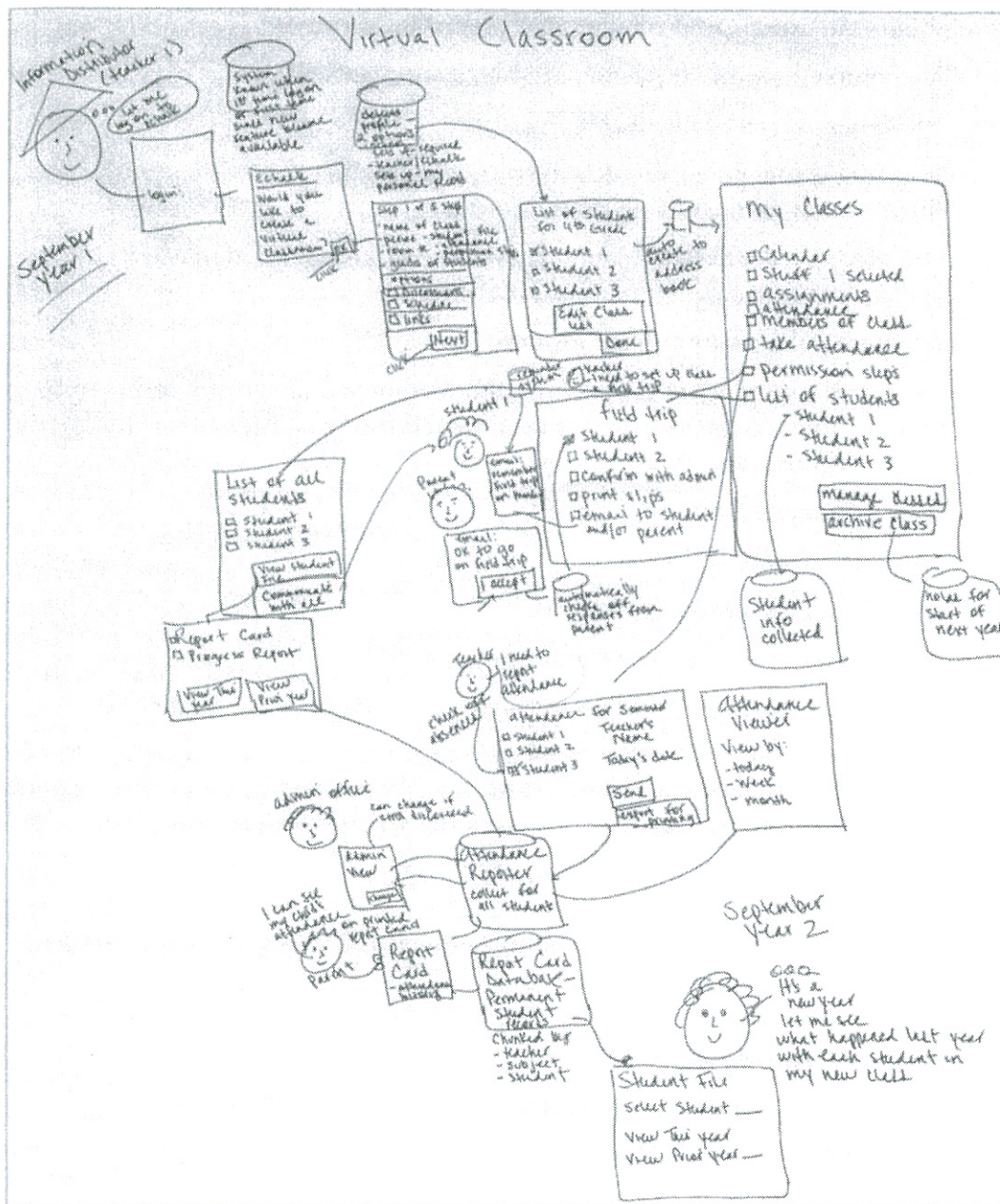


Figure 11-5:
The eChalk virtual classroom vision shows a high-level view of the new work practice. The details are worked out in storyboarding.

The team reviews each vision and then together lists all of the vision's positive attributes (pluses) and then all of the vision's negative attributes (minuses). Listing the positive before the negative forces the team to really look at the vision and not just react to the pieces each individual doesn't like. It ensures that the value in the vision is not lost.

As the team lists the minuses, they should be thinking of design ideas to overcome these problems. Team members then write their design ideas on a Post-it® note, and post them on the vision at the end of the evaluation.

If you have split up into more than one group to create your visions you will need to get back together and share all the new visions. This sharing can be done right before the visions are evaluated.

Pluses represent aspects of the design that work. Pluses are:

- It supports the work of the user as given in the data collected.
- The design is technically doable.
- It has good sales potential—it will create a good value proposition for the sales force to pitch to the buying decision-makers.
- This design is doable within our organization—the organization will be able to collaborate or manage its processes to deliver it.
- It supports the organizational mission.

Negatives are the opposite of the positives, but do not include missing issues, roles, or situations. A vision can be evaluated only on what it tried to address, not on holes it did not address. Minuses are:

- This design doesn't support the work.
- It isn't technically doable.
- This design has low sales potential.
- It isn't consistent with our organization mission.
- It requires too much organizational politics or new processes to succeed.

Sometimes people will be at the extremes of how they feel about the vision: they can love it or hate it. Tell the team that if anyone loves the vision, they must come up with at least three minuses. If they hate the vision, they must come up with at least three pluses.



Example: eChalk

Here are some of the pluses and minuses the eChalk team identified for their Online Classroom vision:

Pluses

- Shares student progress
- Organizes information by class and student
- Has different views of data
- Links to teacher home page
- Lets teacher distribute different information
- Gets teachers to do all of this—work practice is being supported
- Gets information to parent
- Gives teachers control over their information management
- Empowers the teachers, gives them more control—levels the playing field

Minuses

- Will this really save time, or is it more work?
- Need to get buy-in from the school administrators
- Are there concerns that student information is too readily accessible?
- Still have to have manual ways to communicate with parents and students who don't have access to technology.

Create one consolidated vision

After the team has finished creating and sharing the multiple visions they should create one high-level, consolidated vision. The consolidated vision combines visions to eliminate the negatives and reinforce the positives. It presents all the core elements into one vision on one page so that the team has a clear focus for their design work.

To create a consolidated vision, consider the following:

- If the visions do not conflict, draw a new vision, combining parts from each as needed.
- If the visions are actually working out multiple cases in a process, abstract out the overall work flow and core concepts and represent them on one page. Keep the other pages for backup.
- If the visions conflict because they operate from differing business assumptions or technology platforms, agree to the technology or business assumption from which you will operate. Then circle the parts of each vision that will work within those assumptions. You will be surprised how many elements of a conflicting vision can be quickly reconceptualized to fit within the chosen assumptions. Then draw one picture representing the new direction.
- If there are parts of the vision that need technical investigation before they can be considered viable, note them and start a technical team investigating. Design assuming that the technology is stable until otherwise notified. Identify a simplified way to achieve the same intent and storyboard that also.

A consolidated vision can be constructed by two people in a half a day. You may ask why it takes so long. Although it is a “redrawing” of what you already have, it is also an opportunity to clarify what you have drawn and create a coherent representation of what you will be designing. A good consolidated vision guides the team throughout the subsequent design process and keeps them from expanding the scope of the project.

When you are done, produce a clean drawing of the final vision on a flipchart or put it online. Consider writing a short narrative of the vision to ensure that you remember all the elements. This is your very high-level use case or user story.

Initial visions are working documents to help the team make their design conversations concrete. These visions do not really need to be put online. Once you have a consolidated vision that you want to use for designing the system, you want to put it online to help communicate to the rest of your organization. Use a drawing/graphics program such as Visio®, PowerPoint®, or CorelDRAW®.

Hang the consolidated vision on the wall in your team room. If appropriate, communicate your vision to stakeholders or management by walking them through the drawing or create a series of slides introducing the product or system concepts (see the box, **Communication out: Visions create a shared understanding across the organization**).

Example: Shopping

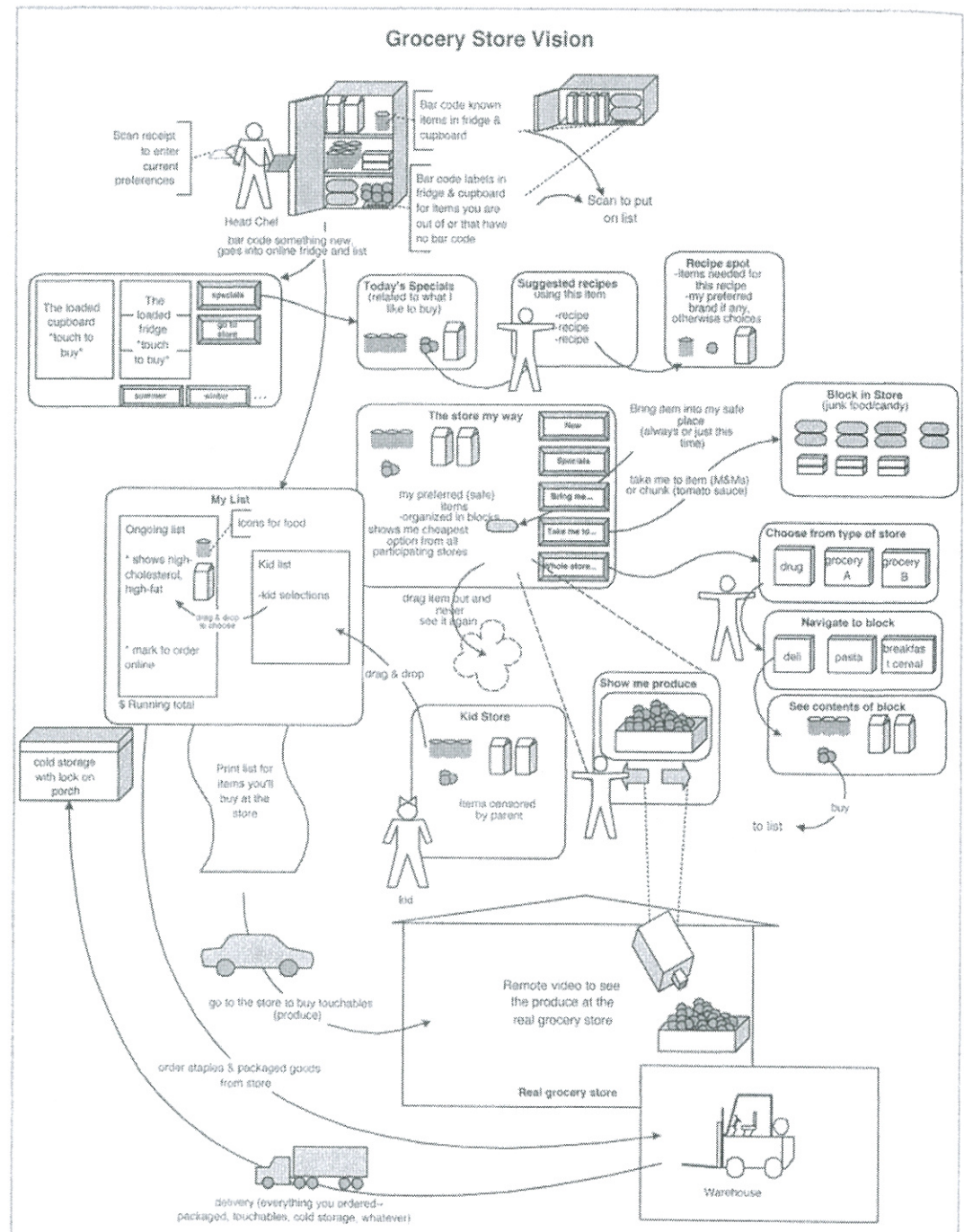
This is the vision for Online Shopping. It is an at-home system using a flat panel or tablet and a hand-held input device for making selections, entering text, and reading



bar code labels. The household member can go to the refrigerator or other location where the panel is stored, and then move around the house with it.

Figure 11-6 is an online form of a vision that can be stored and annotated to remember what the team was thinking. The picture of the vision is supported by a narrative that describes it. Consider videotaping a narrative (see example) to save the vision as one of our clients did when they had to stop a project before it was complete. Having a written description or videotape record is very useful to help you remember the specific content of the vision if time is going to pass before you actually work out the details.

Figure 11-6:
Shopping vision
picture that goes
with the support-
ing narrative.



Example vision narrative: Shopping

This is the vision for Online Shopping. It is an at-home system using a flat panel or tablet and handheld device to make selections, enter text, and read bar code labels. The household member can go to the refrigerator or other location where the panel is stored, and then move around the house with it.

The underlying system stores information from prior purchases to create this household's buying patterns and preferences, and uses this data to populate what is displayed on the panel's screen. One way to add items to a shopping list that appears on the flat panel is to swipe printed bar code labels that are in the refrigerator, cupboards, and other locations of the house.

Another way to add items to the list is by viewing "virtual" household places and "virtual" stores and making selections from them. A virtual household place, for example a virtual cupboard, allows the customer to select individual items, shelves, or entire cupboards or refrigerators and add them to the list. The virtual store's contents—aisles, blocks, or departments; individual items and brands; seasonal items; and store specials—are displayed and organized the way this specific shopper approaches shopping, and displays only the preferred items, brands, and so on. Other household members can have their own special virtual stores, such as a Kid Store that contains only items that have been preapproved by the parents.

Recipes are linked to items. Multiple stores participate, so virtual household and store contents can come from any store and the shopper may opt to see which store has the best price for a given item or only to see items based on price. The bricks-and-mortar store has a video camera that displays real-time images on the shopper's in-home panel so the customer can see what produce or meat actually looks like before adding them to the list. Items are added to the list by touching the image. Items are permanently removed from the household place or store by dragging them out.

There are two ways to place an order and take delivery. The customer can create the list, order online, and then items are delivered to the home. A cold storage box with a lock at the residence is used when the customer is not at home for the delivery. Customers also can assemble their list at home, print it, and then go to the store to do the actual purchasing.

**Communication out: Visions create shared understanding across the organization**

Products usually do not slip because of technology. Products slip because organizations cannot settle on the real requirements and design early enough to operate smoothly. Design from customer data helps this because organizations can look to the data for guidance instead of arguing with each other over who knows the customer best. The vision, produced by a cross-functional team immersed in customer data, creates a shared direction for the product or system. Visioning creates the context for hearing and synthesizing individual design ideas into a coherent business direction.

The more complex the product, the more complex the organizational buy-in process. Everyone wants information and everyone wants their opinion considered. If you are in a large corporation, you will need to get buy-in for your design from other parts of the organization. The vision session provides an excellent opportunity to bring in divergent views, create organizational buy-in and at the same time communicate your work.

For example, one of our clients held five different visioning sessions across their multinational company, bringing in key stakeholders to walk the walls and brainstorm. The team found that everybody was inventing the same thing. Just as work practice is circumscribed by its context, when people from the same company, designing the same kind of technology, looking at the same customer data, considering the same business case, go through the visioning process, the visions overlap from group to group. But the team now had fabulous buy-in from across the corporation!

